Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Scott, Bryan B.S.		
Project Title:	The Evaluation and Selection of Catalyst to Mineralize Organic Impurities in Reclaimed Water		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	To date 2 catalysts have been tested and a third is currently underway. Current results are most promising. Testing began on September 21, 2003. The synthetic carbon substrate catalyst developed by Environmental Life Support Technologies (ELS) is effectively oxidizing ISS ersatz feed to TOC levels less that analyzer detection limits (less than 50 ppb TOC) at 90 deg. C. Feed ersatz was 11 ppm TOC at 25 ml/min (43 min residence time). The ELS catalyst is a 'synthetic carbon' based catalyst loaded with 5% Platinum and Ruthenium. The substrate is 30 mesh resin bead that has been 'fired' at a high temperature. ELS technologies is the manufacturer and previous testing in a 10cc reactor by Cliff Jolly at ELS labs has shown good activity and lifetime. JSC's new alloy20 reactors are 1.9 liters so flow ranges will hopefully be widespread enough for good kinetic data. The Engelhard catalyst evaluation was promising but loss of activity was observed after a few weeks of testing. Early data indicates the catalyst ESCAT 226 will oxidize a 10 ppm TOC Ersatz feed stream to levels below the 500ppb		

Task Description:	standard using air with temperatures ranging from 95 C to 130 C, but catalyst slowly becomes leass active with time until the target TOC cannot be reached. This conclusion was based on repetitive runs with identical test conditions. Inspection of the catalyst beads showed erosion of the alumina beads and therefore loss of Platinum. Catalyst surface scans show a high load of iron coming off the reactor walls onto the catalyst. New reactors were fabricated from alloy20 for future runs. The second catalyst tested showed no visible degradation probably due to it's monolithic design. The ceramic honeycomb cylinders are coated with platinum but the surface area and adsorption characteristics are not as high as a metal oxide bead. Best TOC reduction observed was 65% at minimum flow and 130 deg C on feeds ranging from 2 ppm to 20 ppm TOC. The work to date is the result of ongoing improvements in equipment, analysis and procedures initiated in August of 2002. (replacement of the feed pump, fabrication and installation of new reactors larger and corrosion resistant, etc.)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New technologies in wastewater reclamation are beneficial to all water purification processes in any environment.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)